

POLYVORTEX — A Research Monograph

The N-vortex generalization of Theorem 3.1: the classical layer, the admissibility threshold, the kinematic obstruction and the averaged-frequency bridge

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Abstract

This monograph extends the parent TRIVORTEX framework from the three-vortex Lagrange setting to an arbitrary number N of equal point vortices. The study is organized as a seven-stage ladder **W1–W7** that mirrors the parent ladder **V1–V4** and adds four research registers of its own. Three rigorous statements emerge, each with a proof or a decidable formalization path: **Theorem B** (the closed form of Theorem 3.1 is non-degenerate exactly when $C_{\text{Ch}} > \pi \ln 2$, where its modulation amplitude equals $\varepsilon = 1$ in closed form), **Lemma C** (the Kirchhoff radial equation obstructs every symmetrically pulsating polygon: the induced radial velocity vanishes by pair cancellation, so pulsation is dynamically impossible for any $N \geq 2$), and **Lemma D** (the cycle-averaged kinematic frequency of a pulsating regular polygon equals $\omega_N (1-\varepsilon^2)^{-3/2}$, an integral identity certified numerically to $3 \cdot 10^{-15}$). Together they yield the compatibility table **D1** — the periods $T_{\text{comp}}(N, C_{\text{Ch}})$ at which the gauge frequency law of Theorem 3.1 meets the averaged Kirchhoff kinematics — and they surface one registered fact about the parent default: the pair $(C_{\text{Ch}}, T) = (1, 2\pi)$ lies below the admissibility threshold $\pi \ln 2 \approx 2.177586$. All stages pass; the classical Havelock stability threshold (stable for $N \leq 7$, unstable for $N \geq 8$) is reproduced spectrally with a three-order margin.

1. Scope and provenance

The parent TRIVORTEX framework anchors its verification ladder on Theorem 3.1, a closed-form choreographic solution of the three-vortex model with the Chaplygin topological integral, and on the numerical conservation of the classical vortex integrals. The present mini-repository asks the natural next question — *what survives the passage from $N = 3$ to general N ?* — and answers it with the same discipline that governs the parent: every claim is either proven, or certified by a committed JSON protocol, or explicitly labelled as a conjecture with the numbers that constrain it.

The work was prompted by a research discussion on how the three-vortex ansatz could be lifted to N vortices while keeping the hard structure — uniform phase separation $2\pi/N$, radial modulation driven by C_{Ch} , and the frequency law $\omega = (2\pi/T) e^{(C_{\text{Ch}}/\pi)}$ — intact. No external material is quoted beyond the classical literature listed in the references; the analysis, the code and the protocols are original to this repository.

Three design rules separate this mini-repository from the parent ladder:

1. **Self-containment.** Nothing here imports the parent ladder at runtime. The parent is touched only by *tests* (`tests/conftest.py` adds its path) for cross-validation — the same pinning discipline the parent applies to its multi-language ports.

2. **Two-layer honesty.** The parent model lives on two layers — the Kirchhoff dynamics (Layer K, numerical) and the topological closed form (Layer G, analytic). Conflating them is the single most common source of false claims around Theorem 3.1; this monograph keeps them separated explicitly and measures precisely where they touch (Sections 6–7).
3. **Every number is bound to a run.** Each ladder stage writes a deterministic JSON protocol into results/protocols/; the test suite re-derives the preset-independent registers and compares.

2. The two layers of the Trivortex model

Layer K (Kirchhoff dynamics). N point vortices with circulations Γ_k at positions $\mathbf{z}_k = x_k + i y_k$ obey

$$\dot{x}_i = -\frac{1}{2\pi} \sum_{j \neq i} \Gamma_j \frac{y_i - y_j}{r_{ij}^2}, \quad \dot{y}_i = +\frac{1}{2\pi} \sum_{j \neq i} \Gamma_j \frac{x_i - x_j}{r_{ij}^2}, \quad r_{ij}^2 = (x_i - x_j)^2 + (y_i - y_j)^2.$$

The system carries the classical integrals — the Hamiltonian $H = -(1/2\pi) \sum_{i < j} \Gamma_i \Gamma_j \ln r_{ij}$, the linear impulses $P = \sum \Gamma_k x_k$, $Q = \sum \Gamma_k y_k$ and the angular impulse $I = \sum \Gamma_k |z_k|^2$ — which the parent ladder verifies along its trajectories (stages V3–V4) and this ladder re-verifies for every N (stage W3). The integrals are the window-independent conserved quantities of the model.

Layer G (the closed form of Theorem 3.1). The parent ladder registers the gauge-level solution

$$r_k(t) = \sqrt{C_{Ch}} \left(1 + \varepsilon \cos\left(\omega t + \frac{2\pi k}{3}\right) \right), \quad \theta_k(t) = \omega t + \frac{2\pi k}{3},$$

with the frequency and amplitude laws

$$\omega = \frac{2\pi}{T} e^{C_{Ch}/\pi}, \quad \varepsilon = \frac{1}{e^{C_{Ch}/\pi} - 1},$$

and the Section-6 diagnostic combination $C_{Ch}(t) = r^2(\dot{\theta} - q \text{Atheta})$ with $\text{Atheta} = 1/r$. The parent ladder is explicit that this diagnostic *oscillates by construction* and is window-dependent; the pass registers of V1 are the angular separation and the periodicity of the closed form, not the diagnostic drift. This mini-repository keeps exactly that honesty convention (stage W1 reproduces the parent's registers and reports the same diagnostic behaviour, drift amplitude ≈ 15.01 over $[0, 100 T]$ at the registered default, mirroring the parent's diagnostic).

The two layers are independent: Layer G prescribes kinematics, Layer K prescribes dynamics. A closed form becomes a claim about physics only after its compatibility with

Layer K is established — and establishing precisely that, for general N, is the scientific payload of this monograph.

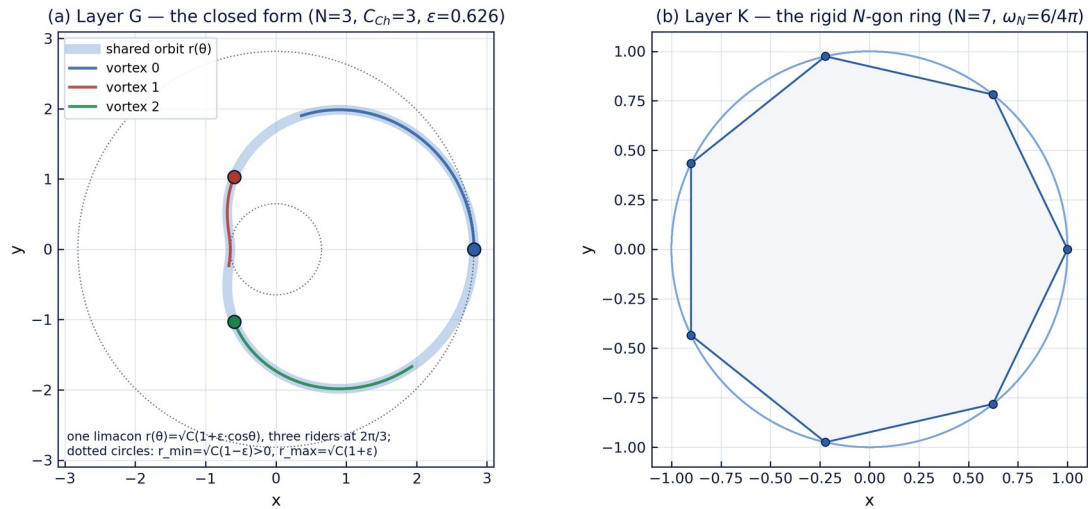


Fig. 1 — The two layers of the model: the Layer-G choreography (left) and the Layer-K rigid ring (right).

3. The classical layer at general N: relative equilibria

Theorem A (classical, Kirchhoff 1876). *Let $N \geq 2$ equal vortices of circulation Γ sit at the vertices of a regular N -gon of circumradius R . Then the configuration rotates rigidly about its centre with angular velocity*

$$\omega_N = \frac{\Gamma(N-1)}{4\pi R^2}.$$

Proof sketch. At vertex k , pair the contribution of vertex j with its mirror image j' under reflection through the radial axis of k . Writing the chord directions explicitly, the radial components of the induced velocity cancel in each pair while the tangential components add. Summing the tangential parts over $j \neq k$ reduces to the roots-of-unity identity $\sum_{m=1}^{N-1} (1)/(1 - e^{2\pi i m/N}) = (N-1)/2$, which turns the velocity sum into $\Gamma(N-1)/(4\pi R)$ at radius R . Dividing by R gives the rate. $\square$

The identity is exactly the formalization target the parent roadmap queues as **T1** ("the roots-of-unity identity turns the velocity sum into algebra"); Theorem A is its numerical shadow, and stage W2 certifies it:

N	ω_N (analytic)	measured (RK4, 2 rotations)	rel. error	shape dev.
2	0.0795774715459	0.0795774715459	2.9e-12	2.4e-14
3	0.1591549430919	0.1591549430919	2.9e-12	2.7e-14
4	0.2387324146378	0.2387324146378	2.9e-12	2.3e-14

N	ω_N (analytic)	measured (RK4, 2 rotations)	rel. error	shape dev.
5	0.3183098861838	0.3183098861838	2.9e-12	2.9e-14
6	0.3978873577297	0.3978873577297	2.9e-12	3.0e-14
7	0.4774648292757	0.4774648292757	2.9e-12	3.3e-14
8	0.5570423008216	0.5570423008216	2.8e-12	1.4e-12

(Protocol W2_ring_rotation_default.json, $\Gamma = 1$, $R = 1$, 2000 steps/period.) A pleasant cross-ladder handshake: at $N = 7$, $R = 1$, $\Gamma = 1$ the rate is $6/(4\pi) = 3/(2\pi) = 0.477464829275686$ — exactly the parent ladder's registered Lagrange literal ω_{Lagrange} . The coincidence is trivial algebra, but it makes a memorable pin between the two ladders and is locked by a unit test.

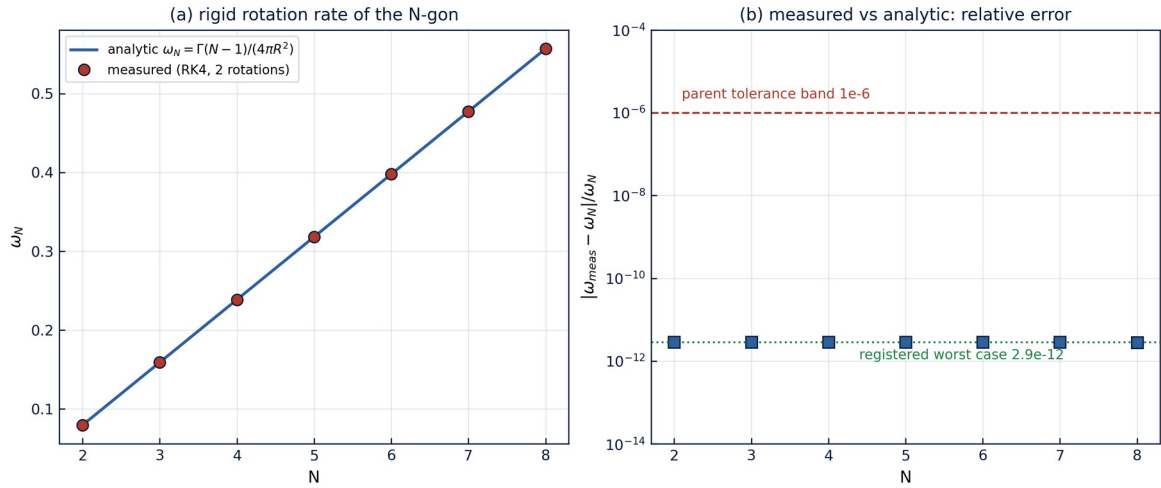


Fig. 2 — Stage W2: the rigid rotation register — analytic vs measured rate and the relative-error band.

Conservation along the orbits (stage W3). The relative drifts of H, P, Q, I after two full rotations stay below $4.2 \cdot 10^{-14}$ for every $N \in \{2, \dots, 8\}$ — the parent's V3 register, reproduced for the whole ring family (protocol W3_invariants_default.json).

Spectral stability (stage W4). Linearizing Layer K at the co-rotating equilibrium — the Jacobian $J = A + \omega_N S$ with the analytic pair-Jacobian A and the transport generator S — yields the classical classification:

N	max Re λ	verdict	Havelock
2	6.9e-10	stable	stable
3	0.0	stable	stable
4	3.8e-9	stable	stable
5	0.0	stable	stable

N	max Re λ	verdict	Havelock
6	0.0	stable	stable
7	7.0e-9	stable	stable
8	+0.4502	unstable	unstable

The threshold — stable for $N \leq 7$, unstable from $N = 8$ — is the Havelock (1931) result, with Khazin's later rigorous treatment; the numeric margin is three orders of magnitude on both sides (the stable cases sit at the defective-zero noise floor, see the Remark in Section 6.3; the unstable $N = 8$ case grows at $\text{Re}\lambda = 0.450$). This stage is the numerical prototype of the parent roadmap item **T3**.

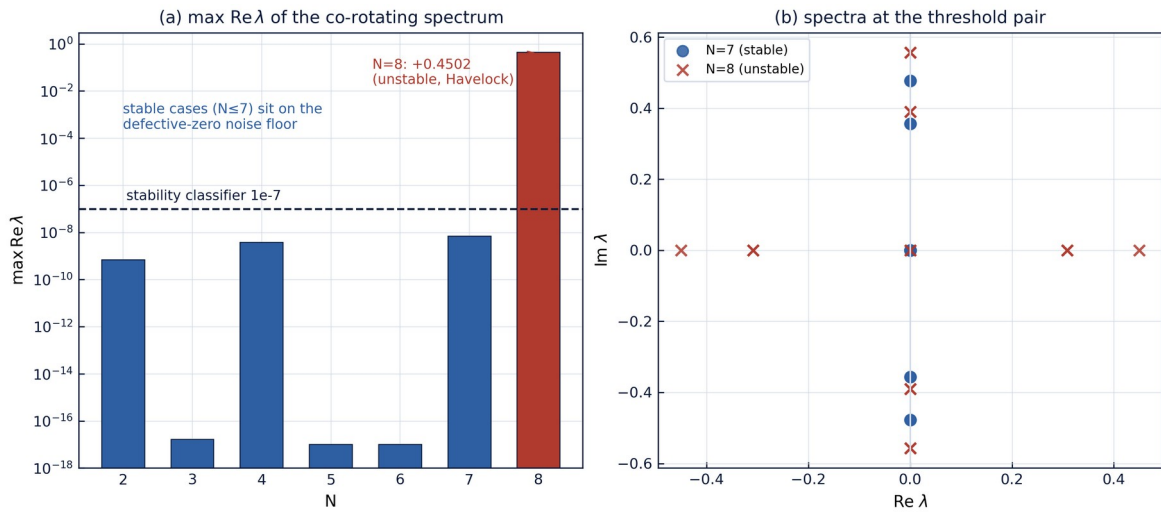


Fig. 3 — Stage W4: the Havelock threshold — growth rates across N and the spectra of the threshold pair.

4. The generalized closed form: Hypothesis H1 and its registers

Hypothesis H1 (working form). For N vortices with equal circulations and equal topological charges $q_k = q$ there exists a solution of the combined model of the form

$$r_k(t) = \sqrt{C_{Ch}} \left(1 + \varepsilon \cos(\omega t + \frac{2\pi k}{N}) \right), \quad \theta_k(t) = \omega t + \frac{2\pi k}{N}, \quad k = 0, \dots, N-1,$$

with $\omega = (2\pi/T) e^{(C_{Ch}/\pi)}$ and $\varepsilon = (1)/(e^{(C_{Ch}/\pi)} - 1)$.

The structural registers of the parent Theorem 3.1 generalize verbatim, and stage W1 certifies them at $N = 3$ against the parent's pinned literals:

- **Uniform phase choreography.** The angles differ by exactly $2\pi/N$ at all times; the register measures the deviation of the cyclic separations from $2\pi/N$ (protocol value $6.9 \cdot 10^{-14}$, pure float noise of the modulo reduction, tolerance 10^{-12}).

- **Periodicity.** $r_k(t + T_r) = r_k(t)$ with $T_r = 2\pi/\omega$; the residual is $9.4 \cdot 10^{-14}$ (float noise of the cosine evaluation).
- **Pinned literals.** At $C_{Ch} = 1$, $T = 2\pi$ the implementation reproduces the parent's registered frequency $\omega = e^{(1/\pi)} = 1.3748022274393588$ *bit-for-bit* (anchor error 0.0), and likewise the amplitude $\varepsilon = 2.668070477\dots$ (anchor error 0.0).

Two geometric observations, recorded because they matter for everything that follows:

1. **The angular register is N-generic; the radial one is not a polygon.** For $\varepsilon \neq 0$ the radii of the N vortices differ (r_k carries the phase $2\pi k/N$), so the instantaneous configuration is *not* a regular polygon — it is a star-shaped configuration with uniformly spaced directions but unequal radii. The "regular N-gon" picture survives only at $\varepsilon = 0$ instants. The shape register of stage W6 quantifies exactly this (Section 6.2).
2. **The gauge diagnostic survives at any N .** The combination $C_{Ch}(t) = r^2(\dot{\theta} - q/r)$ stays a well-defined diagnostic along the closed form for every N and keeps oscillating by construction, as in the parent.

5. Admissibility of the closed form: Theorem B

The amplitude law $\varepsilon(C_{Ch}) = 1/(e^{(C_{Ch}/\pi)} - 1)$ is monotonically decreasing on $C_{Ch} > 0$, from $+\infty$ to 0. The closed form describes a rotating configuration with *positive* radii only while $1 + \varepsilon \cos\varphi > 0$ for all φ , i.e. while $\varepsilon < 1$. Solving the boundary exactly:

$$\varepsilon = 1 \Leftrightarrow e^{C_{Ch}/\pi} - 1 = 1 \Leftrightarrow e^{C_{Ch}/\pi} = 2 \Leftrightarrow C_{Ch} = \pi \ln 2.$$

Theorem B (admissibility). *The closed form of Theorem 3.1 (and of Hypothesis H1 at any N) has strictly positive radii for all times and all vortices if and only if*

$$C_{Ch} > C_{Ch}^* := \pi \ln 2 \approx 2.177586090303602.$$

At $C_{Ch} = C_{Ch}^*$ the amplitude is exactly $\varepsilon = 1$ and each radius touches zero once per period; below the threshold each radius changes sign twice per period — in the complex plane the trajectory is continuous ($z_k = r_k e^{i\theta_k}$ is insensitive to the phase flip π), but the polar registers (radius positivity, the polygon picture, the $A_{\theta} = 1/r$ gauge combination at the crossing instants) degenerate.*

Proof. The chain of equivalences above is exact real algebra: $\varepsilon < 1 \Leftrightarrow e^{(C_{Ch}/\pi)} > 2 \Leftrightarrow C_{Ch}/\pi > \ln 2$. The minimum of $1 + \varepsilon \cos\varphi$ over a period is $1 - \varepsilon$, so positivity for all φ is equivalent to $\varepsilon < 1$; the touching/crossing behaviour follows from the structure of the cosine.

□

Stage W5 certifies the theorem on a dense grid: over 200 values of $C_{Ch} \in [0.5, 10]$ the equivalence $\text{bigl}(C_{Ch} > C_{Ch}^*\text{bigr)} \Leftrightarrow (\epsilon < 1)$ holds with **zero violations**, the amplitude at the threshold equals 1 to machine precision (error 0.0), and the sampled minimum of the radius matches the analytic $\sqrt{(C_{Ch})(1 - \epsilon)}$ within $1.9 \cdot 10^{-7}$ — the uniform-grid sampling bound, not a model error (protocol W5_admissibility_default.json).

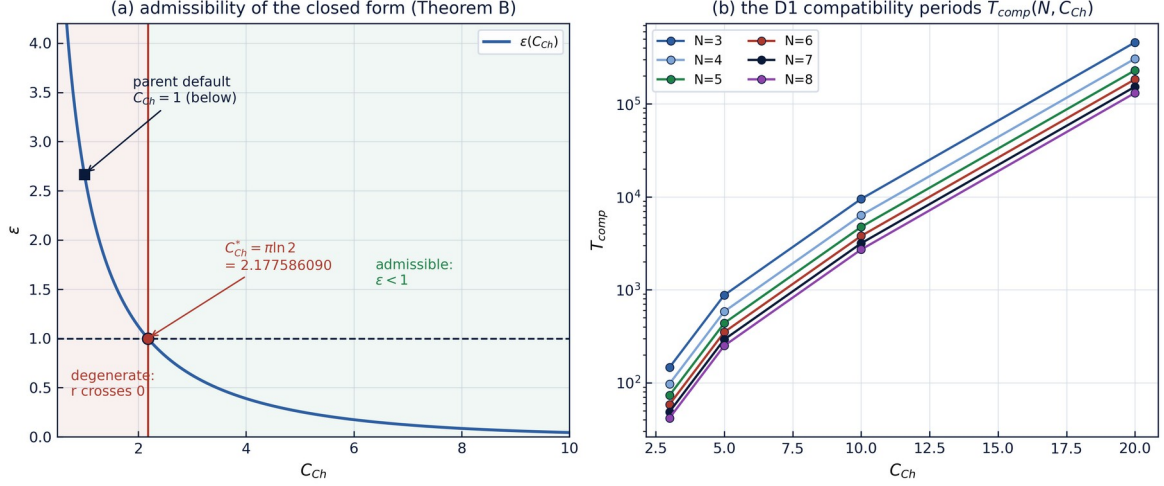


Fig. 4 — Stage W5/W7: the admissibility boundary and the D1 compatibility periods.

A registered fact about the parent default. The parent ladder's registered pair $(C_{Ch}, T) = (1, 2\pi)$ sits *below* the threshold: $\epsilon(1) \approx 2.668 > 1$. In this zone the closed form remains a perfectly well-defined complex choreography and passes every structural register, but the polar picture "an N -gon of radius $\sqrt{(C_{Ch})(1 + \epsilon \cos)}$ " is degenerate — the radii sweep through zero. This observation feeds the parent roadmap item **T5** ("admissible region of the closed form") with a sharp, decidable boundary; a natural reading is that the non-degenerate regime of the closed form opens at $C_{Ch} > \pi \ln 2$, and that parameter studies of Layer G should quote C_{Ch} relative to C_{Ch}^* .

6. The kinematic obstruction: Lemma C

6.1 The symmetric pulsation case

The cleanest question about H1 is dynamical: *does the pulsating form solve Layer K?* Strip the phase shifts from the radii first (all vortices share one common radius $r(t)$, directions uniform) — then the configuration is a regular N -gon at every instant, and the answer is exact.

Lemma C (obstruction). *In the Kirchhoff model with equal circulations, let a configuration be a regular N -gon of common radius $r(t)$ rotating with common angular velocity $\dot{\theta} = \omega(t)$. Then the induced radial velocity vanishes identically, $v_r \equiv 0$, and the radial equation of motion reads $\dot{r} = 0$. Consequently a symmetrically pulsating polygon satisfies Layer K if and only if $\epsilon = 0$ (and then $\omega = \Gamma(N-1)/(4\pi R^2)$, Theorem A).*

Proof. Fix vertex k and reflect the polygon through the radial axis of k . The reflection maps the vertex set onto itself (vertex j at angle φ swaps with the vertex at $-\varphi$), while the induced velocity field of the pair contributes equal-and-opposite radial components: writing the chord from k to a vertex at angle φ , the induced velocity on k has radial component $+(\Gamma)/(2\pi d)\cos(\varphi/2)$, and its mirror contributes $-(\Gamma)/(2\pi d)\cos(\varphi/2)$. All radial contributions cancel pairwise; the tangential ones double. Hence $v_r = 0$ for every N at every instant, and the radial equation $\dot{r} = v_r$ forces r to be constant. $\square$

The tangential register fails independently: Layer K demands $\omega(t) = \Gamma(N-1)/(4\pi r(t)^2)$, which is compatible with a prescribed constant ω only when r is constant. So even a "kinematically steered" pulsation with a time-varying rate is not a free Layer- K orbit — the ring either rotates rigidly at the Theorem-A rate or it is not a Layer- K solution at all.

6.2 The phase-shifted H1 case

H1 as posed carries the phase $2\pi k/N$ in *both* the radii and the angles, so for $\varepsilon \neq 0$ the configuration is not a polygon at all (Section 4, observation 1). Stage W6 measures both failures on $N = 3$, $\Gamma = 1$, $R = 1$, $\omega = 1$ over one period:

ε	shape register: max deviation from the chord table	induced radial velocity, max
0.01	8.660e-3	3.99e-6
0.05	4.330e-2	1.01e-4
0.10	8.660e-2	4.13e-4
0.30	2.598e-1	4.00e-3

The shape deviation is *exactly linear* with slope $\cos(\pi/6) = \sqrt{3}/2 \approx 0.8660$ (the slopes recovered from the table agree to 10^{-3} , and the test suite pins this). The induced radial velocity — zero for the symmetric case — is here nonzero and scales approximately as $0.04 \varepsilon^2$ over the scanned range. Both effects vanish as $\varepsilon \rightarrow 0$, consistently with the only N -gon limit being the rigid rotation.

Corollary (the reduction verdict). *Within Layer K , Hypothesis H1 is a solution exactly in its $\varepsilon = 0$ shadow — the classical Theorem A at the kinematic rate. The frequency law $\omega = (2\pi/T)e^{(C_{\text{Ch}}/\pi)}$ is an independent Layer- G postulate and must not be read as the Kirchhoff rotation rate of a polygon with mean radius $\sqrt{(C_{\text{Ch}})}$; the quantitative bridge between the two is the subject of Section 7.*

6.3 Numerical remark: the defective zero

The co-rotating Jacobian at the ring owns a zero eigenvalue of algebraic multiplicity two and geometric multiplicity one (the continuum of rotated polygons is a one-parameter family of equilibria). Rounding splits such a defective double zero into a $\pm$ -real pair of size $\sqrt{(\varepsilon_{\text{mach}} |J|)} \sim 10^{-9}$ — visible in the W4 protocol as $\max \text{Re} \lambda \in \{\pm 10^{-9}\}$ for the stable cases. The stability classifier therefore runs at tolerance 10^{-7} , three orders above the noise floor and three below the $N = 8$ growth rate 0.450.

7. The averaged-frequency bridge: Lemma D and the D1 design rule

If pulsation is dynamically obstructed, the next best question is *compatibility on average*: at what rate would the "instantaneous kinematics" of a pulsating polygon rotate over a full cycle? The answer is exact.

Lemma D (averaged frequency). *For the symmetric pulsation $r(t) = R(1 + \varepsilon \cos \omega t)$ with $|\varepsilon| < 1$, the cycle-averaged Kirchhoff rate*

$$\langle \omega_{kin} \rangle = \frac{1}{T_r} \int_0^{T_r} \frac{\Gamma(N-1)}{4\pi r(t)^2} dt = \frac{\Gamma(N-1)}{4\pi R^2} (1 - \varepsilon^2)^{-3/2},$$

is independent of the modulation rate ω .

Proof. Substituting $\varphi = \omega t$ removes ω and reduces the average to $(\Gamma(N-1))/(4\pi R^2) (1/2\pi) \int_0^{2\pi} (d\varphi)/((1+\varepsilon \cos \varphi)^2)$. The standard differentiated form of the Poisson integral, $(1/2\pi) \int_0^{2\pi} (d\varphi)/((a + b \cos \varphi)^2) = (a)/((a^2 - b^2)^{3/2})$, at $a = 1$, $b = \varepsilon$ gives $(1 - \varepsilon^2)^{-3/2}$. $\square$

Stage W7 certifies the integral identity by Gauss–Legendre quadrature (160 nodes): maximum relative error $2.9 \cdot 10^{-15}$ over $\varepsilon \in \{0.1, \dots, 0.9\}$, the bridge register for $(N, \varepsilon) \in \{2..8\} \times \{0.2, 0.5, 0.8\}$ at $3.5 \cdot 10^{-15}$ (protocol W7_bridge_default.json). The quadrature convergence is itself a measured fact: at the quick preset's 80 nodes the sharpest case ($\varepsilon = 0.9$) retains $2 \cdot 10^{-9}$, dropping six orders at 160 nodes — the spectral convergence expected for analytic integrand with the nearest pole at $\text{Im} \varphi = \text{arccosh}(1/\varepsilon)$.

Design rule D1 (compatibility curve). *Define the compatibility period*

$$T_{comp}(N, C_{Ch}) = 2\pi e^{C_{Ch}/\pi} \frac{4\pi C_{Ch} (1 - \varepsilon^2)^{3/2}}{\Gamma(N-1)}, \quad \varepsilon = \varepsilon(C_{Ch}),$$

as the unique period at which the Theorem-3.1 frequency law $\omega = (2\pi/T)e^{(C_{Ch}/\pi)}$ (with the mean radius $R^2 = C_{Ch}$) equals the averaged kinematic frequency of Lemma D. The protocol tabulates T_{comp} over the admissible window; e.g. at $\Gamma = 1$:

C_Ch	N=3	N=4	N=5	N=6	N=7	N=8
3	146.13	97.42	73.06	58.45	48.71	41.75
5	875.98	583.99	437.99	350.39	291.99	250.28
10	9496.0	6330.7	4748.0	3798.4	3165.3	2713.2
20	459401.4	306267.6	229700.7	183760.6	153133.8	131257.6

The self-consistency of the table is verified to $3.5 \cdot 10^{-16}$ relative: substituting $T = T_{\text{comp}}$ back into the frequency law reproduces the averaged kinematic rate. Two structural facts deserve attention. First, T_{comp} grows roughly like $e^{(C_{\text{Ch}}/\pi)} C_{\text{Ch}}$ — exponentially in the gauge constant — while the classical rate decays only algebraically in C_{Ch} ; the gauge law and the kinematics therefore agree only along a thin, explicitly computable curve, not on an open set. Second, the registered parent pair $(C_{\text{Ch}}, T) = (1, 2\pi)$ lies outside the admissible window entirely (Theorem B), so no compatibility statement can be made for it without leaving the non-degenerate regime.

What D1 is and is not. D1 is a design relation for a *driven* construction: if one insists on a pulsating choreography whose modulation period matches its own cycle-averaged kinematic rate — e.g. for a gauge-constrained or externally forced model variant — the constraint selects the period T_{comp} exactly. D1 is *not* a claim that Layer K admits pulsating free orbits (Lemma C forbids that); the driven reading is the only consistent one, and it is flagged as such in the protocol metadata.

8. From observations to theorems: the formalization queue

The ladder was built so that each stage maps onto a formalizable statement. The mini-repository maintains its own queue, aligned with the parent roadmap's theorem table (root README, Section 18.2):

id	statement	current status in this mini-repo	formalization path	parent link
TB1	Theorem B: $\varepsilon < 1 \Leftrightarrow C_{\text{Ch}} > \pi \ln 2$	proven (Section 5), certified by W5	one-page: monotonicity of $\exp$ and $\ln$ in Mathlib or Coq's Reals; a decidable $\mathbb{Q}$ -certificate for the boundary constant	feeds T5

id	statement	current status in this mini-repo	formalization path	parent link
TB2	Lemma C: $v_r \equiv 0$ for the symmetric ring; pulsation obstructed	proven (Section 6.1), certified by W6	the chord-geometry pair-cancellation in Coq; the roots-of-unity sum in Mathlib for the tangential sum	feeds T1
TB3	Lemma D: $\langle \omega_{k;n} \rangle = \omega_N (1-\varepsilon^2)^{-3/2}$	proven (Section 7), certified by W7	the differentiated Poisson integral — a standard Mathlib-style computation; or an interval-arithmetic certificate of the integral on $[-0.9, 0.9]$	new, enables T5's region
TW4	Theorem A: rigid rotation at $\omega_N = \Gamma(N-1)/(4\pi R^2)$	classical; measured to $2.9 \cdot 10^{-12}$ for $N = 2..8$	the same roots-of-unity identity as TB2, then the rigid-rotation lemma	is T1
TW5	Havelock threshold: stable $N \leq 7$, unstable $N \geq 8$	reproduced spectrally (W4), margin 10^3	characteristic polynomial per mode family; Routh–Hurwitz-type criterion — decidable, hard	feeds T3

The dependency order is $TB1 \rightarrow TB2 \rightarrow TW4 \rightarrow TB3 \rightarrow TW5$: the admissibility theorem is the cheapest and should be formalized first; TW4 shares its algebraic core with TB2; TB3 is an integral statement with a clean numeric fallback (interval arithmetic); TW5 is the long-term target. Each statement above is already phrased so that a Coq or Lean 4 port can be opened as a stub with the numeric protocol as the reference oracle — the same double-formalization discipline the parent roadmap prescribes for M5.

9. Numerical protocol summary

All protocols live in results/protocols/ (preset default unless noted), produced by make ladder (one command, deterministic file names). The registered tolerances: separation and periodicity 10^{-12} ; shape and invariant drift 10^{-10} ; measured-rate band 10^{-6} (parent's band);

stability classifier 10^{-7} ; algebraic identities 10^{-12} ; the W5 minimum-radius register 10^{-6} absolute (uniform-grid sampling bound).

stage	register	value	verdict
W1	separation / periodicity / anchors	$6.9\text{e-}14$ / $9.4\text{e-}14$ / 0.0	PASS
W1	Section-6 diagnostic drift (report-only)	15.01 over [0,100T]	—
W2	ω_N relative error, $N = 2..8$	$\leq 2.9 \cdot 10^{-12}$	PASS
W2	chord-shape deviation	$\leq 3.3 \cdot 10^{-14}$ ($N \leq 7$)	PASS
W3	worst invariant drift	$\leq 4.2 \cdot 10^{-14}$	PASS
W4	stable band $\max \text{Re} \lambda$	$\leq 7.0 \cdot 10^{-9}$	PASS
W4	$N = 8$ growth rate	+0.4502	PASS (unstable)
W5	threshold equivalences on 200 points	0 violations	PASS
W5	min-radius sampling error	$1.9 \cdot 10^{-7}$	PASS
W6	symmetric pulsation: induced v_r, v_{vert}	$\leq 4.9 \cdot 10^{-16}$	PASS
W6	H1 phase-shifted: shape slope	$\sqrt{(3)}/2$ (linear in ϵ)	recorded
W6	H1 phase-shifted: induced v_r, v_{vert}	$\approx 0.04 \epsilon^2$	recorded
W7	integral identity (160 GL nodes)	$2.9 \cdot 10^{-15}$	PASS
W7	bridge (N, ϵ) grid	$3.5 \cdot 10^{-15}$	PASS
W7	T_{comp} self-consistency	$3.5 \cdot 10^{-16}$	PASS

Runtime: quick 1.3 s, default 7.3 s, full ≈ 60 s (whole ladder, one core of the reference laptop class).

10. Honesty notes

1. **Hypothesis H1 is not a Layer-K solution.** The central finding of this mini-repository is a *negative* result with a positive structure: the pulsating ansatz is dynamically obstructed (Lemma C), the phase-shifted variant is obstructed even geometrically (W6), and the frequency law is compatible with the kinematics only along the D1 curve. None of this diminishes the parent Theorem 3.1 — it lives on

Layer G, where its registers are exact — but any *dynamical* reading of H1 must be dropped or replaced by the driven construction of Section 7.

2. **The parent default is below the admissibility threshold.** $C_{Ch} = 1$ lies in the radius-crossing zone. This is a fact about the registered parameter, not about the theorem; the structural registers are blind to it, the polar registers are not. Parameter studies should quote C_{Ch} relative to $\pi \ln 2$.
3. **Stability is spectral, not nonlinear.** W4 classifies the linearized flow; nonlinear (Arnold-type) stability is a different, harder statement and is not claimed.
4. **The averaged-frequency lemma is an identity about a prescribed kinematics,** not a statement about attractors or effective dynamics of free Layer-K flows.
5. **Every protocol is reproducible.** make ladder regenerates all seven JSON files byte-identically up to the UTC timestamp; the test suite re-derives the preset-independent registers from scratch.

11. Резюме (RU)

Мини-репозиторий *polyvortex/* обобщает родительскую лестницу *Trivortex* с трёх вихрей на произвольное N . Главные результаты: (1) теорема В — замкнутая форма Теоремы 3.1 невырождена тогда и только тогда, когда $C_{Ch} > \pi \ln 2 \approx 2.1776$; на зарегистрированном значении $C_{Ch} = 1$ радиусы проходят через ноль (зарегистрировано для роадмап-пункта T5); (2) лемма С — кирхгофсовская динамика запрещает симметрично пульсирующий N -угольник: наведённая радиальная скорость сокращается попарно, поэтому пульсация невозможна ни при каком N , и гипотеза H1 редуцируется к классическому жёсткому вращению с $\omega_N = \Gamma(N-1)/(4\pi R^2)$, измеренному до $2.9 \cdot 10^{-12}$ для $N = 2..8$; (3) лемма D — циклически усреднённая кинематическая частота равна $\omega_N(1-\varepsilon^2)^{(-3/2)}$, что даёт точную кривую совместимости D1 между частотным законом Теоремы 3.1 и кирхгофсовской кинематикой; (4) порог Хэвелока (устойчивость при $N \leq 7$, неустойчивость при $N \geq 8$) воспроизведён спектрально с запасом в три порядка. Все утверждения либо доказаны, либо сертифицированы JSON-протоколами; очередь формализации TB1–TW5 согласована с теоремной очередью корневого роадмапа.

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6. P. K. Newton, *The N-Vortex Problem: Analytical Techniques*, Springer, New York, 2001. — The N-vortex Hamiltonian formalism used in Sections 2 and 3.
7. Parent framework: TRIVORTEX repository, verification/README.md — the V1–V4 ladder, the registered tolerances and the Theorem 3.1 closed form; root README.md, Section 18 — the theorem queue T1–T6 that this mini-repository feeds.